Task 4 – Report

Documents

- **Enrollment folder**: contains 30x5 signatures data (30 writers, 5 true signatures each writer).
- **Verification folder**: contains 30x45 signatures data (30 writers, 45 genuine / forgeries signatures).
- **gt.txt file**: contains 30x45 lines in the shape "001-02 g", which means "writer 1, verification signature 2, true".
- **users.txt file**: contains the list of the 30 writer numbers.

Steps

1. **Normalized feature extraction**

   1.1 After loading the data, for each writer we create

   - A list containing the 5 true signatures corresponding the writer (enrN)
   - A list containing the 45 unclassified signatures (verN)

   1.2 Then we extract the features x, y, the pressure and we calculate Δx, Δy and then normalize the 5 features.

2. **Calculation of the best distance for each writer using DTW**

   2.1 We chose to take a starting value for the distance matching of 30. If the distance between two signatures is less than this value, it is considered as the same signature (= the same writer).

   2.2 We use the DTW algorithm between the 5 true signatures of the first writer and all the 45 unclassified signatures. During the process, if there is a match, check if it is a TP or a FP. If there is no match, check if it is a TN or a FN. At the end of the calculations, we calculate the precision, the recall and the f-score.

   2.3 We iterate the DTW algorithm on 20 epochs. In each iteration, we reduce the distance value by 1.

   2.4 After the 20 epochs, we make a graph and find the best value considering precision, recall and f-score for each writer.

3. **DTW algorithm used as in point 2 to find each writer's best distance.**

3.1 For the 30 writers one after the other, we use the DTW algorithm with 30 for the distance. Then, we calculate the precision, recall and f-score for the writer.

4. DTW algorithm used with the best distance of writer 1.

4.1 We calculate recall, precision and f-score. They will be used to compare the use of one writer's best distance applied to the whole set and the generalized distance we will find in step 5.

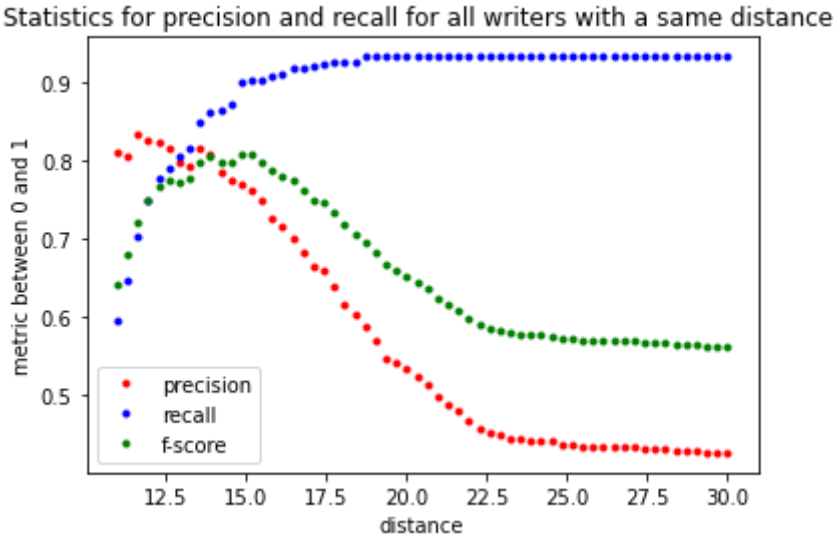
5. DTW algorithm used for 60 epochs to find the best general distance.

5.1 For each epoch, we use a distance in the range of the best individual distances (start at the maximal best value found with the 30 writers and decrease this value until the minimal best value found with the 30 writers). The value used for decreasing is the “step” variable and is calculated as the difference of the maximal and minimal distance, all divided by the number of epochs. We calculate the precision, the recall and the f-score.

5.2 After having calculated these values, we make a graph considering precision, recall and f-score for the whole set of writers with the 60 used distances.

5.3 After, we find the result of the best generalized distance and then, compare the associated results of the best generalized distance applied to the whole set of writers to the best distance of the writer 1 also applied to the whole set of writers.

5.4 With the generalized distance, the recall is really better and the precision a bit worse, so the f-score is really better.



Results discussion

The unique best distance for the generalized distance is : ~14.864407

We calculated the average of the Recall, Precision, and F-score for the whole set of writers with the calculated best distance of the writer 1.

The result was:

Best distance for writer 1 : 11

Average Recall for the whole set of writers: 0.5948888888888888

Average Precision for the whole set of writers: 0.8112698412698413

Average F-score for the whole set of writers: 0.6412776310779775

If we calculate the average of the Recall, Precision, and F-score for the whole set of writers with the **generalized distance** the result is :

Best distance : 14.864407

Average Recall for the whole set of writers: 0.9

Average Precision for the whole set of writers : 0.7703774141235441

Average F-score for the whole set of writers: 0.8091939828433639

Comparing the associated results of the generalized distance with the best distance of writer 1, we can see that with the generalized distance, we have really improved the recall value and a bit decreased the precision, but this association of precision and recall leads to a better f-score, so a better result.

Good results ?

The results are good.

To increase the quality of the results, we can increase the number of epochs to have a better approximation of the best generalized distance.

To increase the results, we can use more features and more complex algorithms.